



OMNIX QUANTUM LTD
DECISION GOVERNANCE INFRASTRUCTURE

UNITED STATES PATENT AND TRADEMARK OFFICE
PROVISIONAL PATENT APPLICATION

TITLE OF INVENTION:

**TRAJECTORY INVARIANT ENGINE FOR AUTOMATED
GOVERNANCE DECISION PIPELINES WITH BOUNDED
EVOLUTION ENFORCEMENT, MULTI-DIMENSIONAL
INVARIANT MONITORING, AND GLOBAL REGIME
COLLAPSE DETECTION**

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Title: TRAJECTORY INVARIANT ENGINE FOR AUTOMATED GOVERNANCE DECISION PIPELINES WITH BOUNDED EVOLUTION ENFORCEMENT, MULTI-DIMENSIONAL INVARIANT MONITORING, AND GLOBAL REGIME COLLAPSE DETECTION

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FIELD OF THE INVENTION

The present invention relates to automated decision governance systems, and more particularly to a trajectory invariant engine that operates downstream of a multi-checkpoint governance pipeline to enforce bounded evolution constraints over sequences of governance decisions, detecting and blocking decision paths that push the system toward globally inadmissible operational regions even when each individual decision in the sequence passes all upstream checkpoint validations independently.

BACKGROUND

I. THE STATE VALIDATION PROBLEM IN SEQUENTIAL DECISION GOVERNANCE

Multi-checkpoint governance pipelines for automated decision systems evaluate each proposed decision against a set of validation criteria to determine whether the decision is admissible. These pipelines typically assess the current state of the system and the proposed decision in isolation: each decision is evaluated against thresholds for signal quality, probabilistic confidence, risk exposure, coherence, and compliance. A decision that passes all checkpoints is approved for execution.

This architecture has a fundamental architectural limitation that existing systems have not addressed: it validates states but not paths.

The path-state distinction is well understood in formal verification and control theory but has not been applied to automated governance decision pipelines. A system that validates only states can reach a valid state via an invalid path — and may reach a globally inadmissible region through a sequence of locally valid transitions.

Concrete failure mode: Consider a governance pipeline where each individual decision correctly passes all 8 checkpoints with adequate signal quality, probability scores, and risk metrics. However, the sequence of decisions over a 20-cycle window shows: (1) monotonically increasing risk exposure across all approved decisions; (2) probability scores hovering just above the minimum threshold for 4 consecutive cycles; (3) simultaneous degradation in signal coherence across 3 or more independent assets. Each individual decision is locally valid. The trajectory as a whole is a leading indicator of systemic failure — the system is walking toward a cliff one valid step at a time.

Existing governance architectures fail to detect this pattern because:

1.1 No Trajectory Memory: Checkpoint pipelines are stateless with respect to decision history. Each evaluation is independent of previous evaluations.

1.2 No Cross-Asset Correlation: Individual-asset governance does not detect simultaneous degradation across multiple assets as a single systemic event requiring escalated response.

1.3 No Path Constraints: Systems that validate only current-state admissibility have no mechanism to block a locally valid decision that continues a globally inadmissible trajectory.

1.4 Reactive Rather Than Predictive: Existing governance systems respond to violations after they are measurable in current state. They do not detect trajectory trends that predict imminent violations before they manifest in individual-decision metrics.

The present invention addresses all four inadequacies through the Trajectory Invariant Engine (TIE).

SUMMARY OF THE INVENTION

The present invention provides a Trajectory Invariant Engine (TIE) that operates as a post-pipeline gate in automated governance decision systems. The TIE:

- Maintains a rolling trajectory state history for each operational asset or domain, persisted across decision cycles.
- Evaluates five trajectory-level invariants over the historical window after each decision passes all upstream checkpoint validations, converting APPROVED decisions to HOLD when blocking invariants are violated.
- Enforces bounded evolution: the system must not only reach valid states — it must reach them via valid paths.
- Detects global regime collapse — simultaneous degradation across a configurable minimum number of independent assets — and escalates the response accordingly.
- Operates in configurable fail mode: by default, any TIE module failure produces a pass-through result that does not block the upstream pipeline decision (fail-open); in critical-domain deployments, TIE may be configured to fail-closed, treating any module failure as a blocking HOLD to prevent ungoverned decisions in high-stakes environments.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

I. ARCHITECTURAL POSITION AND DESIGN PRINCIPLE

The Trajectory Invariant Engine occupies a specific architectural position in the governance pipeline:

```
Signals → N-checkpoint pipeline → [APPROVED] → TIE → APPROVED | HOLD  
[BLOCKED] → TIE bypassed
```

Critical design principle: TIE only evaluates decisions that have already been APPROVED by the full upstream checkpoint pipeline. BLOCKED decisions are never modified by TIE — they are already blocked for independent reasons. TIE's function is to catch the category of decisions that pass all individual checkpoints but whose approval would continue an inadmissible trajectory.

This asymmetric application is architecturally significant: TIE is not an additional checkpoint that a decision must pass. It is a trajectory-level veto that can convert an APPROVED decision to HOLD based on the accumulated history of prior approved decisions.

The bounded evolution principle: A compliant governance system must not only reach valid states — it must reach them via valid paths. TIE enforces that approved decisions do not push the system toward globally inadmissible regions, even when each individual decision appears valid in isolation. This principle distinguishes TIE from all existing checkpoint-based governance architectures.

The TIE is applicable to any domain in which sequential automated decisions must maintain trajectory-level safety properties, including but not limited to: algorithmic trading governance, clinical decision support audit, autonomous robotic action sequencing, energy grid dispatch governance, and any regulated domain where regulatory compliance is evaluated

over sequences of decisions rather than individual decisions.

II. TRAJECTORY STATE REPRESENTATION

For each operational asset or domain, the TIE maintains a trajectory state record comprising the following fields recorded at each decision cycle:

- **asset:** Identifier of the asset or domain (e.g., trading symbol, patient ID, robot unit ID)
- **domain:** Operational domain classifier (e.g., "trading", "credit", "medical", "energy")
- **decision:** The governance outcome of the decision (e.g., "APPROVED", "BLOCKED", "HOLD")
- **probability_score:** Composite probability confidence score from the upstream pipeline [0–100]
- **risk_exposure:** Quantified risk exposure metric from the upstream pipeline [0–100]
- **signal_coherence:** Signal coherence metric from the upstream pipeline [0–100]
- **trend_persistence:** Measure of trend continuity in the signal [0–100]
- **stress_resilience:** Measure of system resilience to adverse conditions [0–100]
- **logic_consistency:** Measure of internal decision logic consistency [0–100]

Trajectory states are stored in a persistent database with a configurable rolling window (in one embodiment: the most recent 20 states per asset). In one embodiment, states are stored in a PostgreSQL `trajectory_states` table. In other embodiments, any persistent storage mechanism may be used, including in-memory storage for deployments where cross-restart persistence is not required.

III. FIVE TRAJECTORY INVARIANTS

The TIE enforces five trajectory-level invariants, each with a configurable window size and threshold. Invariants are classified as HOLD (blocking) or WARNING (non-blocking but logged).

III.A. Invariant 1 — *RISK_MONOTONIC_ASCENT (HOLD)*

Definition: If `risk_exposure` exceeds a configurable threshold (in one embodiment: 70 out of 100) for a configurable number of consecutive decisions (in one embodiment: 5 consecutive decisions), TIE converts the next APPROVED decision to HOLD.

Rationale: A monotonically ascending risk exposure trajectory over consecutive approved decisions is a leading indicator that the system is incrementally increasing its exposure to a level that would be inadmissible if reached in a single decision, but which is individually admissible at each step. No individual decision violates the risk threshold; the trajectory as a whole represents a ratchet toward excessive risk.

Detection algorithm: The engine inspects the N most recent trajectory states for the asset. If all N states have `risk_exposure` above the threshold, the invariant is violated.

In one embodiment: N = 5, threshold = 70. In other embodiments, N and the threshold are configurable parameters.

III.B. Invariant 2 — *PROBABILITY_DEAD_ZONE (HOLD)*

Definition: If `probability_score` remains below a configurable threshold (in one embodiment: 35 out of 100) for a configurable number of consecutive decisions (in one embodiment: 4 consecutive decisions), TIE converts the next APPROVED decision to HOLD.

Rationale: A probability score that persistently hovers just above the pipeline's minimum approval threshold for multiple consecutive cycles indicates that the system is operating in a region of marginal confidence. Each individual decision marginally passes; the trajectory reveals that the system has been operating at the edge of admissibility for an extended period, which is a different risk from a single decision at marginal confidence.

Detection algorithm: The engine inspects the N most recent trajectory states. If all N states have `probability_score` below the threshold, the invariant is violated.

In one embodiment: N = 4, threshold = 35. In other embodiments, N and the threshold are configurable parameters.

III.C. Invariant 3 — COHERENCE_STRUCTURAL_DECAY (HOLD)

Definition: If `signal_coherence` remains below a configurable threshold (in one embodiment: 40 out of 100) for a configurable number of consecutive decisions (in one embodiment: 3 consecutive decisions), TIE converts the next APPROVED decision to HOLD.

Rationale: Signal coherence measures the internal consistency of the input signals to the governance pipeline. A sustained decay in signal coherence across consecutive decisions indicates structural degradation in the information environment — the signals that the pipeline is using to make decisions are becoming progressively less reliable. The short window (in one embodiment: 3 decisions) reflects the severity of coherence decay as an early warning of data quality failure.

Detection algorithm: The engine inspects the N most recent trajectory states. If all N states have `signal_coherence` below the threshold, the invariant is violated.

In one embodiment: N = 3, threshold = 40. In other embodiments, N and the threshold are configurable parameters.

III.D. Invariant 4 — TRAJECTORY_VOLATILITY (WARNING only)

Definition: If the standard deviation of `probability_score` over a configurable window (in one embodiment: 8 decisions) exceeds a configurable threshold (in one embodiment: 32), TIE emits a WARNING without blocking the decision.

Rationale: High variance in the probability score across the trajectory window indicates that the system is oscillating between high-confidence and low-confidence decisions without stabilizing. This pattern is diagnostic of an unstable signal environment and warrants monitoring, but its severity is insufficient on its own to block an otherwise approved decision. The WARNING is logged and included in the TIE result for downstream monitoring and audit purposes.

Detection algorithm: The engine computes the standard deviation of `probability_score` over the N most recent trajectory states and compares against the threshold.

In one embodiment: N = 8, threshold = 32. In other embodiments, N and the threshold are configurable parameters.

III.E. Invariant 5 — GLOBAL_REGIME_COLLAPSE (HOLD with escalation)

Definition: If a configurable minimum number of independent assets or domains (in one embodiment: 3 or more) simultaneously have `probability_score` below the `PROBABILITY_DEAD_ZONE` threshold in their most recent trajectory state, TIE detects a global regime collapse and applies escalated response (HOLD on all affected assets, with high-severity logging).

Rationale: Simultaneous degradation across multiple independent assets is qualitatively different from per-asset degradation. When three or more independent governance streams simultaneously exhibit low-probability conditions, this indicates a systemic event affecting the broader environment in which the governance system operates — not individual asset-specific noise. The cross-asset correlation of degradation is not detectable by per-asset checkpoint validation and requires the trajectory history of multiple assets to be analyzed jointly.

Detection algorithm: For each asset with a trajectory state in the current evaluation window, the engine retrieves the most recent state and checks whether `probability_score` is below the `DEAD_ZONE_THRESHOLD`. If the count of such assets equals or exceeds `GLOBAL_COLLAPSE_ASSET_MIN`, a global collapse is declared. This check requires database access; in memory-only deployments, global collapse detection is disabled.

In one embodiment: `GLOBAL_COLLAPSE_ASSET_MIN` = 3, threshold = `DEAD_ZONE_THRESHOLD` = 35. In other embodiments, these parameters are configurable.

IV. TRAJECTORY SCORE

In addition to individual invariant evaluation, TIE computes a composite trajectory health score (in one embodiment: ranging from 0 to 100) for the asset at the time of evaluation. The score reflects the aggregate health of the trajectory across all evaluated dimensions.

The trajectory score serves two purposes:

- It provides a continuous signal for monitoring dashboards and governance metrics systems, even when no invariants are violated.

- It enables configurable warning thresholds that allow operators to receive early warning of trajectory degradation before any blocking invariant is triggered.

V. CONFIGURABLE FAIL MODE — FAIL-OPEN AND FAIL-CLOSED

The TIE supports two configurable failure modes to accommodate deployments with different risk profiles:

Fail-Open (default): Any exception, database connectivity failure, or internal error within the TIE produces a pass-through result in which the upstream pipeline's APPROVED decision is returned unchanged. The pass-through reason is recorded in the TIE result for audit purposes. This mode ensures that TIE's failure mode is identical to its absence — the governance pipeline continues to operate as if TIE were not present, preserving the safety properties of the upstream pipeline without adding new failure modes. Fail-open is the recommended default for domains where decision continuity is preferred over conservative blocking.

Fail-Closed (critical domains): In high-stakes deployment environments — including but not limited to clinical decision support, autonomous systems with physical actuation, energy grid dispatch, and defense-grade governance — TIE may be configured to fail-closed (in one embodiment: via `TIE_FAIL_MODE=closed` environment variable). In fail-closed mode, any TIE module failure, database unavailability, or internal exception produces a HOLD result rather than a pass-through. This ensures that no APPROVED decision proceeds without a successful trajectory invariant evaluation in domains where an ungoverned decision carries unacceptable risk.

The choice between fail-open and fail-closed is a deployment configuration parameter, not an architectural constraint. Both modes record the fail-event with its reason in the TIE result for full audit traceability.

VI. CONFIGURATION AND ENABLE/DISABLE

The TIE is configurable via a deployment environment variable (in one embodiment: `TIE_ENABLED`). When disabled, TIE returns a pass-through result for all decisions without evaluating any invariants. This enables staged deployment in environments where trajectory history must be accumulated before invariant evaluation begins.

VII. ALTERNATIVE EMBODIMENTS

7.1 Extensible Invariant Registry: The five invariants described above represent one embodiment. The invariant set is explicitly designed to be extensible: new invariants may be registered at deployment time without modifying the core TIE engine. Each invariant is defined by a window size, a threshold, a severity classification (HOLD or WARNING), and an evaluation function over the trajectory state history. In other embodiments, additional trajectory-level invariants may be defined and registered, including but not limited to: monotonic decline in `trend_persistence` across K consecutive decisions; joint threshold violations across correlated pairs of metrics (e.g., simultaneous decline in `logic_consistency` and `signal_coherence`); time-weighted decay invariants that apply exponentially decaying weights to historical states; domain-specific invariants defined by regulatory mandate (e.g., maximum consecutive approvals in a single asset class for financial governance, or maximum consecutive actuations for robotic governance); and cross-domain invariants that correlate trajectory metrics across multiple operational domains simultaneously.

7.2 Configurable Invariant Severity: In one embodiment, the severity classification of each invariant (HOLD vs. WARNING) is configurable per deployment, allowing operators to run invariants in observation-only mode before enabling their blocking behavior.

7.3 Cross-Domain Application: The TIE architecture is domain-agnostic. The five invariants and the bounded evolution principle apply equally to: clinical decision support (patient treatment trajectory), autonomous robotic action governance (robot operational trajectory), energy grid dispatch (grid stability trajectory), credit underwriting pipelines (portfolio risk trajectory), and any domain where sequential decisions must maintain trajectory-level safety properties.

7.4 Weighted History: In one embodiment, trajectory states are weighted by recency — more recent states contribute more to invariant evaluation than older states — enabling faster detection of recent trajectory changes while maintaining sensitivity to longer-term trends.

CLAIMS

Claim 1 (Broad — Core TIE) — A computer-implemented trajectory invariant engine for automated governance decision pipelines, comprising: a trajectory state store that maintains a rolling history of governance decision outcomes and associated metrics for each operational entity across decision cycles; a trajectory invariant evaluator that analyzes said rolling history to determine whether a sequence of individually valid governance decisions constitutes a globally inadmissible trajectory; and a trajectory veto mechanism that converts an approved governance decision to a hold decision when trajectory invariants are violated, without modifying blocked decisions.

Claim 2 (Broad — Bounded Evolution) — The system of Claim 1, implementing a bounded evolution principle wherein the system enforces that approved governance decisions must be reached via valid decision paths, not merely that each decision constitutes a valid state in isolation, whereby a locally valid decision that continues a globally inadmissible trajectory is blocked.

Claim 3 (Specific — Risk Monotonic Ascent) — The system of Claim 1, wherein, in one embodiment, a first trajectory invariant detects monotonically ascending risk exposure across a configurable number of consecutive approved decisions above a configurable threshold, and wherein violation of said invariant converts the next approved decision to hold.

Claim 4 (Specific — Probability Dead Zone) — The system of Claim 1, wherein, in one embodiment, a second trajectory invariant detects persistent probability scores below a configurable threshold across a configurable number of consecutive decisions, indicating sustained operation in a region of marginal governance confidence.

Claim 5 (Specific — Coherence Structural Decay) — The system of Claim 1, wherein, in one embodiment, a third trajectory invariant detects structural decay in signal coherence across a configurable number of consecutive decisions, indicating progressive degradation of the information environment in which governance decisions are made.

Claim 6 (Broad — Global Regime Collapse) — The system of Claim 1, further comprising a cross-entity global regime collapse detector that simultaneously analyzes trajectory states across a plurality of independent operational entities, and wherein simultaneous degradation of trajectory metrics across a configurable minimum number of independent entities triggers an escalated governance response distinct from per-entity invariant violations.

Claim 7 (Broad — Trajectory Score) — The system of Claim 1, further comprising a trajectory health score module that computes a continuous composite score reflecting the aggregate health of the decision trajectory across all evaluated dimensions, enabling monitoring and early warning prior to blocking invariant violations.

Claim 8 (Broad — Configurable Fail Mode) — The system of Claim 1, wherein the trajectory invariant engine supports a configurable failure mode comprising: a fail-open mode in which any operational failure produces a pass-through result returning the upstream pipeline decision unchanged; and a fail-closed mode in which any operational failure produces a HOLD result blocking the upstream pipeline decision; wherein the failure mode is selectable per deployment as a configuration parameter to accommodate domains ranging from decision-continuity-preferred to ungoverned-decision-prohibited operational environments.

Claim 9 (Broad — Post-Pipeline Gate) — The system of Claim 1, wherein the trajectory invariant engine is positioned exclusively downstream of the full upstream governance checkpoint pipeline and evaluates only decisions that have been approved by all upstream checkpoints, preserving all upstream blocking decisions without modification.

Claim 10 (Broad — Method) — A computer-implemented method for bounded evolution enforcement in automated governance decision pipelines, comprising: recording trajectory state metrics for each governance decision outcome; evaluating a plurality of trajectory-level invariants over a rolling history of recorded states; detecting globally inadmissible trajectory patterns that would not be detectable by independent evaluation of any individual decision; and converting approved decisions to hold decisions when blocking trajectory invariants are violated.

ABSTRACT

A trajectory invariant engine (TIE) enforces bounded evolution constraints in automated governance decision pipelines by analyzing sequences of individually valid decisions to detect globally inadmissible trajectories. Positioned downstream of a multi-checkpoint governance pipeline, TIE evaluates five trajectory-level invariants over a rolling history of decision outcomes and associated metrics per operational entity: (1) RISK_MONOTONIC_ASCENT detects monotonically increasing risk exposure across consecutive decisions; (2) PROBABILITY_DEAD_ZONE detects persistent marginal-confidence operation; (3) COHERENCE_STRUCTURAL_DECAY detects progressive signal quality degradation; (4) TRAJECTORY_VOLATILITY emits non-blocking warnings for high variance in confidence scores; (5) GLOBAL_REGIME_COLLAPSE detects simultaneous degradation across a configurable minimum number of independent entities as a systemic event. Blocking invariants convert APPROVED decisions to HOLD without modifying BLOCKED decisions. TIE implements the bounded evolution principle: the system must not only reach valid states but must reach them via valid paths. All invariant parameters are configurable; the invariant set is explicitly extensible — new invariants are registered at deployment time without modifying the core engine. TIE supports configurable fail modes: fail-open (pass-through on error, default for continuity-preferred domains) and fail-closed (HOLD on error, for critical domains where ungoverned decisions are prohibited). Cross-entity global collapse detection requires joint analysis of trajectory histories across multiple independent operational entities, not detectable by per-entity checkpoint validation. Applications include algorithmic trading governance, clinical decision support, autonomous robotics, energy grid management, and any regulated domain requiring path-level safety guarantees over sequences of automated decisions. To the inventor's knowledge, this is the first system applying trajectory invariant enforcement with bounded evolution path constraints specifically to automated governance decision pipelines across multiple operational domains.

DRAWINGS

FIG. 1 – TIE Architectural Position

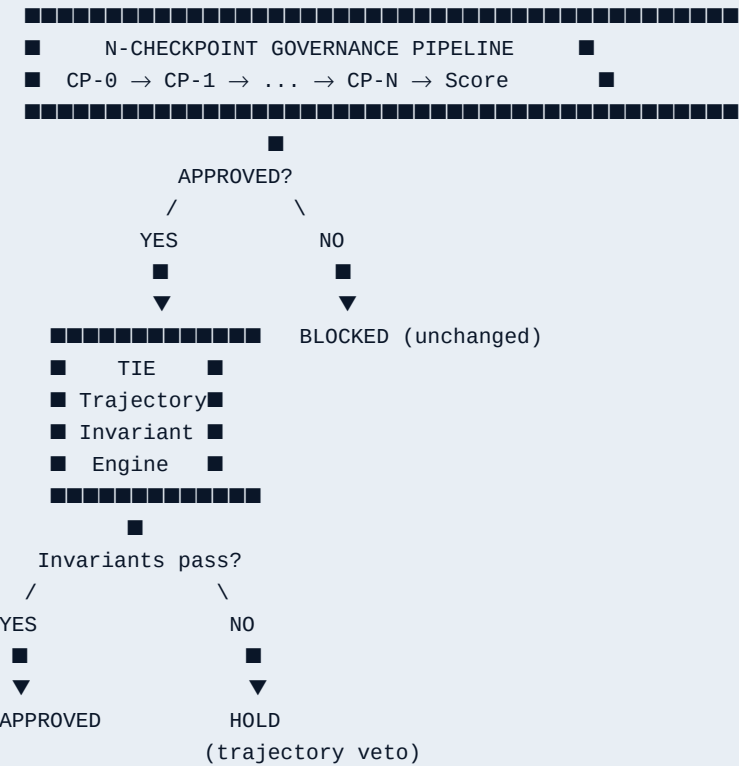


FIG. 2 – Five Trajectory Invariants

Invariant	Window	Threshold	Severity

RISK_MONOTONIC_ASCENT	5	risk > 70	HOLD
PROBABILITY_DEAD_ZONE	4	prob < 35	HOLD
COHERENCE_STRUCTURAL_DECAY	3	coh < 40	HOLD
TRAJECTORY_VOLATILITY	8	σ > 32	WARNING
GLOBAL_REGIME_COLLAPSE	1	≥ 3 assets	HOLD+ESC

All parameters configurable via deployment configuration.

FIG. 3 – Bounded Evolution vs. State Validation

State Validation (existing systems):
Decision N: risk=68 ✓ prob=38 ✓ coh=43 ✓ → APPROVED
Decision N: risk=69 ✓ prob=37 ✓ coh=42 ✓ → APPROVED
Decision N: risk=71 ✓ prob=36 ✓ coh=41 ✓ → APPROVED ← each valid alone

Bounded Evolution (TIE):
Trajectory: risk ascending 5 cycles → RISK_MONOTONIC_ASCENT
 prob in dead zone 4 cycles → PROBABILITY_DEAD_ZONE
Next decision → HOLD (trajectory veto)

FIG. 4 – Global Regime Collapse Detection

Asset A: prob_score = 32 (below 35) ← dead zone
Asset B: prob_score = 29 (below 35) ← dead zone
Asset C: prob_score = 31 (below 35) ← dead zone
Asset D: prob_score = 78 ← healthy

3 assets in dead zone $\geq$ GLOBAL_COLLAPSE_ASSET_MIN (3)
→ GLOBAL_REGIME_COLLAPSE declared
→ HOLD + escalated logging on all affected assets